Handwritten Text Recognition Using TrOCR and EasyOCR: A Comprehensive Comparative Analysis

1st Ilada Geokping Chua 66102010159,
2nd Jinnapat Thongbanguay 66102010232,
3rd Thananya Thanametketkul 66102010240,
4th Renuka Pollok 66102010245
CP461 Introduction to Computer Vision
Advisors: Banphatree Khomkham, PhD and Napa Sae-Bae
Srinakharinwirot University

***Abstract*—Human handwriting possesses high variability and is often difficult to read, especially when characters are disorganized or when images suffer from inconsistent lighting, shadows, and variable quality. These limitations often cause traditional OCR systems to fail in accurately converting handwriting to digital text. Additionally, English handwriting varies significantly across different countries and regions, resulting in character styles that are diverse and unpredictable. Motivated by these challenges, this study aims to develop and evaluate handwriting-recognition models using the GNHK real-world handwriting dataset and to compare the performance of the pre-trained TrOCR, fine-tuned TrOCR, and EasyOCR models. The evaluation explores metrics such as CER, WER, accuracy, and qualitative inspection, offering an in-depth examination of model strengths, weaknesses, and limitations. The results highlight the substantial improvement achieved through fine-tuning and further illustrate the performance gap between Transformer-based and traditional OCR architectures.**

***Index Terms*—Handwritten Text Recognition, TrOCR, EasyOCR, Transformer, CRNN, OCR, GNHK Dataset**

I. Introduction

Human handwriting is inherently diverse due to individual writing habits, cultural differences, and different environmental factors. This variability presents challenges for OCR systems, particularly when dealing with real-world images that contain shadows, inconsistent lighting, blur, noise, or skew. Traditional OCR systems often rely on character segmentation or handcrafted features, which dramatically reduces performance when characters are overlapped or handwritten in cursive styles.

English handwriting also differs widely between global regions, such as Europe, North America, Africa, and Asia. These variations introduce further unpredictability in character shapes and writing conventions. In response to these challenges, this study aims to compare modern OCR methods, such as TrOCR based on Transformers, with widely used classical OCR systems like EasyOCR (CRAFT + CRNN + CTC).

The objective is to analyze how effectively these models handle real-world handwriting, assess the impact of fine-tuning on domain-specific data, and identify strengths, weaknesses, and areas for improvement.

II. Project Objective

The main objectives of this project include:

1) To serve as a Baseline Model for performance comparison against the TrOCR model, which is a modern model based on the Transformer architecture.
2) To evaluate the capability of traditional Deep Learning models (CRAFT + CRNN + CTC) on the task of Handwritten Text Recognition, especially when applied to English handwritten data.
3) Measure the performance difference using evaluation metrics such as CER, WER, and accuracy between a non-specialized model and a model built to handle more complex OCR tasks.
4) To understand the limitations of EasyOCR and analyze areas for improvement compared to a fine-tuned model such as TrOCR.

III. Our Contributions

To achieve these objectives, two major contributions were made:

A. 1) Data Augmentation Pipeline

A hypothesis was formed that expanding the data diversity would improve the model's tolerance to different handwriting variations. Several augmentation techniques were applied:

- Rotation
- Photometric distortions (light adjustments)
- Gaussian noise injection
- Brightness and contrast adjustments

B. 2) Hyperparameter Tuning During Inference

The inference performance of TrOCR was adjusted by using the following:

- `length_penalty = 2.0`: encouraging longer text predictions
- `no_repeat_ngram_size = 3`: preventing repetitive 3-gram sequences

IV. Materials and Methods

A. Dataset Description: GNHK

The GoodNotes Handwriting Kollection (GNHK) dataset was used. It is a large-scale handwriting dataset collected by a professional data-labeling company that contains English handwriting samples from multiple regions.

TABLE I: GNHK Dataset Composition

Region	Images
Europe (EU)	330
North America (NA)	146
Africa (AF)	117
Asia (AS)	94
Total	**687**

B. Data Preprocessing Process

Each image was paired with a JSON annotation that contains text location polygons. Our preprocessing included:

- cropping text segments from raw images,
- generating text labels from JSON,
- applying augmentations only to training data.

C. Model Descriptions

1) TrOCR (Transformer OCR): TrOCR is built on an encoder-decoder Transformer architecture:

- **Vision Transformer Encoder** splits images into patches.
- **Transformer Decoder** generates text autonomously.

Two versions were used:

1) Pre-trained `microsoft/trocr-small-handwritten`
2) Fine-tuned model using the GNHK dataset

2) EasyOCR (CRAFT + CRNN + CTC): EasyOCR comprises:

- **CRAFT**: text detection via character-heatmap estimation
- **CRNN**: feature extraction and sequence modeling
- **CTC Decoder**: translating features into text

V. Experimental Methodology

A. Training and Hyperparameters

For the TrOCR model, we experimented with key hyperparameters during inference, such as setting `length_penalty = 2.0` and `no_repeat_ngram_size = 3` to improve text generation.

B. Evaluation Metrics

We compared the models using Character Error Rate (CER). The results on the validation dataset are shown in Table II.

TABLE II: Comparison of Character Error Rate (CER)

Model	CER (Validation)
TrOCR (Pre-trained)	1.0432
EasyOCR (Pre-trained)	0.6985
TrOCR (Fine-tuned)	**0.2817**

Fig. 1: Pre-trained Model

Fig. 2: Fine-tuned Model

Fig. 3: After using Data Augmentation

Fig. 4: Comparison of TrOCR model outputs showing improvements from pre-training to fine-tuning with augmentation.

TrOCR Performance Analysis: The experimental results highlight a significant improvement in recognition accuracy:

- **Pre-trained Model CER:** The original `microsoft/trocr-small-handwritten` model exhibited a CER of **1.0432** on the GNHK validation dataset.
- **Fine-tuned Model CER:** The fine-tuned model showed significantly better performance, achieving a CER of **0.2817** on the same validation dataset.

Overall, the fine-tuned model demonstrated a remarkable **73.0% reduction** in the Character Error Rate compared to the pre-trained baseline.

C. Experimental Method using EasyOCR

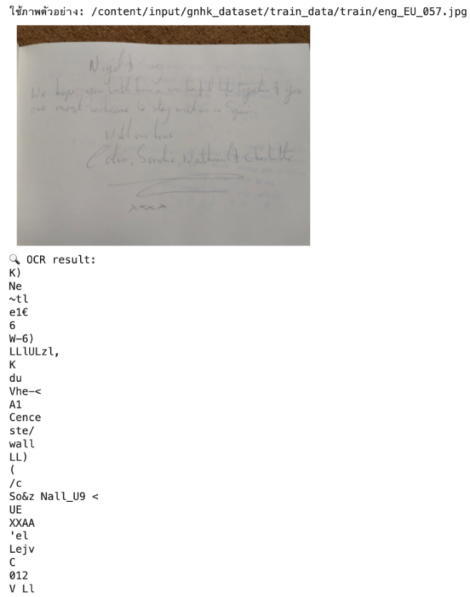

Fig. 5: EasyOCR test result on sample image.

EasyOCR Performance: The pre-trained EasyOCR model exhibited a CER of **0.6985** in the GNHK validation data set. Although accessible, it struggled with the complex handwrit-

ing styles found in the dataset compared to the fine-tuned TrOCR.

VI. Results and Discussion

A. TrOCR: Strengths

- Works well with clear text boundaries
- Handles consistent handwriting styles
- Effective on short words and clean backgrounds

B. TrOCR: Weaknesses

- Struggles with heavy cursive handwriting
- Errors increase with long sequences
- Small visual details (such as dots) are sometimes missed

Performance Details:

Pre-trained Model CER: The original `microsoft/trocr-small-handwritten` model exhibited a CER of 1.0432 on the GNHK validation dataset.

Fine-tuned Model CER: Our fine-tuned model showed significantly better performance, achieving a CER of 0.2817 on the same validation dataset.

Performance Improvement: The fine-tuned model demonstrated a remarkable 73.0% reduction in the Character Error Rate compared to the pre-trained model.

Fine-tuning the `microsoft/trocr-small-handwritten` model on the GNHK dataset resulted in a substantial improvement in handwriting recognition accuracy for this specific domain. This demonstrates the effectiveness of Transfer Learning for adapting pre-trained models to specialized tasks. The fine-tuned model is significantly more accurate at recognizing text within the GNHK dataset than its general-purpose, pre-trained counterpart.

C. EasyOCR: Strengths and Weaknesses

a) Strengths:

- Works well on printed text
- Performs well with clear, non-overlapping characters

b) Weaknesses:

- Not tailored for handwriting
- Cannot be fine-tuned for better performance
- Sensitive to noise, skew, and character overlap

Based on the evaluation metrics (CER, WER, and Accuracy), EasyOCR demonstrated low performance in reading data from the test dataset, especially in cases involving complex text or English handwriting.

- **CER = 69.85%**: This indicates that more than half of all characters were read incorrectly.
- **WER = 88%**: This indicates that almost every word was read incorrectly.
- **Accuracy $\approx$ 34%**: Character accuracy is below 40%.

*** All metrics indicate that EasyOCR cannot read the text in this dataset well and makes a substantial number of errors.

TABLE III: Comparison of Key Features

Feature	TrOCR	EasyOCR
Architecture	ViT + Transformer	CRAFT + CRNN
Context Reasoning	Strong	Weak
Handwriting Support	Very High	Low
Speed	Slower	Faster
Fine-tuning	Supported	Not supported

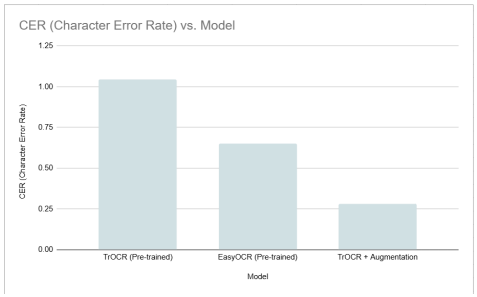

Fig. 6: Comparison CER vs Model

VII. Comparison of TrOCR and EasyOCR

VIII. Conclusion

This project focuses on developing and evaluating an Optical Character Recognition (OCR) system by comparing the performance of two models: EasyOCR and TrOCR. The experiments clearly demonstrate that both models have distinct strengths and limitations, particularly in the context of Handwritten Text Recognition, which is inherently challenging.

From the experiments, EasyOCR exhibited relatively high CER and WER values along with low accuracy, indicating that the model is more suitable for printed text rather than handwritten text. It also struggles to handle various handwriting styles or images that contain noise. In contrast, the pre-trained TrOCR model performed better overall but still produced errors when processing low-quality images or highly complex handwriting. However, the fine-tuned TrOCR model achieved a significantly lower CER, demonstrating the model's ability to learn domain-specific handwriting patterns and improve accuracy substantially when trained with appropriate data.

Throughout this project, the team gained valuable insights into the importance of several key factors, including data quality, preprocessing techniques, model selection, and the use of various evaluation metrics (CER, WER, Accuracy, and qualitative result inspection). Additionally, analyzing the errors made by the models helped us develop a deeper understanding of their behavior and limitations.

IX. References

References

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